

Deterministic Optimization

Large-Scale Optimization

Dantzig-Wolfe
Decomposition

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Dantzig-Wolfe Decomposition 1

Large-Scale Optimization

Dantzig-Wolfe

Learning Objectives for this lesson

- Develop the Dantzig-Wolfe decomposition algorithm for coordination and distributed optimization

Dantzig-Wolfe Decomposition

- Consider a general optimization problem:

$$\begin{array}{ll} (LP) & \min \quad c^\top x \\ & \text{s.t.} \quad \boxed{Dx = b_0} \quad \leftarrow \text{Complicating constraint} \\ & \quad \quad \boxed{\begin{array}{l} Fx = b \\ x \geq 0 \end{array}} \quad \leftarrow \text{Easy constraint} \end{array}$$

- Step 1:** Partition the constraints into complicating constraints and easy constraints.

Dantzig-Wolfe Decomposition

- Rewrite the problem as:

$$\begin{aligned} (LP) \quad & \min \quad \boldsymbol{c}^T \boldsymbol{x} \\ & \text{s.t.} \quad \boldsymbol{D}\boldsymbol{x} = \boldsymbol{b}_0 \\ & \quad \boldsymbol{x} \in P \end{aligned}$$

where $P = \{\boldsymbol{x} : F\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq 0\}$ is assumed to be a *bounded* polyhedron, i.e. a polytope.

- Therefore, P has a finite number of extreme points

Dantzig-Wolfe Decomposition

- Step 2: Use extreme point representation for P
 - Suppose P has N extreme points, denoted as $x^1, x^2, \dots, x^N$
 - Each point x in P can be expressed as a convex combination:

$$x = \sum_{i=1}^N \lambda_i x^i$$

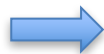
$$\sum_{i=1}^N \lambda_i = 1$$

$$\lambda_i \geq 0, \forall i = 1, \dots, N.$$

Dantzig-Wolfe Decomposition

- Step 3: Substitute extreme point representation to original problem:

$$\mathbf{x} \in P \Leftrightarrow \begin{cases} \mathbf{x} = \sum_{i=1}^N \lambda_i \mathbf{x}^i \\ \sum_{i=1}^N \lambda_i = 1 \\ \lambda_i \geq 0, \forall i = 1, \dots, N. \end{cases}$$



$$(LP) \quad \begin{aligned} \min \quad & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{D}\mathbf{x} = \mathbf{b}_0 \\ & \mathbf{x} \in P \end{aligned}$$

(LP) in $\mathbf{x}$ space



$$\begin{aligned} \min_{\lambda_1, \dots, \lambda_N} \quad & \sum_{i=1}^N \lambda_i (\mathbf{c}^T \mathbf{x}^i) \\ \text{s.t.} \quad & \sum_{i=1}^N \lambda_i (\mathbf{D}\mathbf{x}^i) = \mathbf{b}_0 \\ & \sum_{i=1}^N \lambda_i = 1 \\ & \lambda_i \geq 0, \forall i = 1, \dots, N. \end{aligned}$$



$$\begin{aligned} \min_{\lambda_1, \dots, \lambda_N} \quad & \mathbf{c}^T \left(\sum_{i=1}^N \lambda_i \mathbf{x}^i \right) \\ \text{s.t.} \quad & \mathbf{D} \left(\sum_{i=1}^N \lambda_i \mathbf{x}^i \right) = \mathbf{b}_0 \\ & \sum_{i=1}^N \lambda_i = 1 \\ & \lambda_i \geq 0, \forall i = 1, \dots, N. \end{aligned}$$

Transformed
problem
is in λ space

Summary

- We learned:
 - Three important steps in Dantzig-Wolfe decomposition that transform a complicated optimization problem into another space using extreme point representation of the easy constraints